
Native Geometry vs. Linear Recoverability: Testing Platonic Scaling Across Astronomical Surveys

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Abstract

1 The Platonic Representation Hypothesis predicts increasing representational con-
2 vergence with model scale, but convergence depends on the transformation class
3 under which representations are compared. Across matched Legacy Survey and
4 HSC embeddings from five model families, supervised linear alignment steepens
5 the model-size dependence of cross-survey neighbourhood correspondence relative
6 to native cosine geometry. The amplification occurs in all five families and persists
7 at fixed rank 256, ruling out increased alignment dimensionality from wider em-
8 beddings as the explanation. The fitted maps remain broadly anisotropic, while
9 most of their mKNN gain is carried by a small singular subspace. Sparse SAE
10 and BSF representations provide no advantage over the matched dense-alignment
11 baseline. Thus scaling strengthens linear recoverability more clearly than native
12 geometric similarity.

13 1 Introduction

14 Representations from independently trained models often become more aligned as capability and
15 scale increase [Huh et al., 2024]. The Platonic Representation Hypothesis (PRH) interprets this
16 as convergence toward a shared statistical model of reality. Matched Legacy Survey [Dey et al.,
17 2019]↔HSC [Miyazaki et al., 2018] embeddings provide a comparatively clean testbed: prior work
18 reports increasing mutual k -nearest-neighbour (mKNN) agreement with model capacity [Univer-
19 seTBD et al., 2025]. The central question is not whether any cross-survey signal exists, but whether
20 its *strength* depends on the transformation class under which equivalence is assessed.

21 A substantial literature studies how to *align* independently trained embedding spaces: geometry-
22 preserving orthogonal/Procrustes registration [Mikolov et al., 2013], supervised linear interoperability
23 between representations, and unpaired or pseudo-paired embedding translation [Lample et al., 2018,
24 Moschella et al., 2023, Jha et al., 2025]. Our contribution is not that linear maps can align embeddings.
25 Rather, we ask how the apparent *scaling* of representational convergence changes with the permitted
26 transformation class, and whether the maps that improve recoverability are themselves becoming
27 geometrically simpler.

28 We organize allowed maps as $\mathcal{T}_{\text{id}} \subset \mathcal{T}_{\text{scaled orth}} \subset \mathcal{T}_{\text{affine}}$. At $\mathcal{T}_{\text{id}}$, representations are compared in
29 native coordinates. At $\mathcal{T}_{\text{scaled orth}}$, equivalence is assessed up to rotation/reflection and global scaling
30 ($x \mapsto cQx$ with $Q^\top Q = I$); such maps preserve all pairwise cosines and therefore cosine k NN
31 structure. At $\mathcal{T}_{\text{affine}}$, supervised Ridge may apply anisotropic linear deformation plus translation.
32 Under the primary cosine-mKNN observable, identity and scaled-orthogonal reparameterizations
33 are observationally indistinguishable, since $\cos(cQx_i, cQx_j) = \cos(x_i, x_j)$ for $c > 0$. The exper-
34 imentally meaningful comparison is therefore native cosine geometry (modulo scaled-orthogonal
35 transforms) versus supervised affine-linear recoverability.

On the same held-out objects we measure *native correspondence* $M_{\text{native}} \equiv M_{\text{Dense}}$ (raw cosine mKNN) and *linear recoverability* $M_{\text{linear}} \equiv M_{\text{Dense+Ridge}}$ (mKNN after a supervised global Ridge map fit on paired training objects). The *alignment gain* is $G = M_{\text{linear}} - M_{\text{native}}$. We use “linear recoverability” as shorthand for correspondence exposed by this affine predictor’s linear component (Section 2).

We report three claims. **(1)** Linear recoverability scales more strongly with model size than native correspondence. **(2)** This survives fixed independent-PCA rank 256, so it cannot be explained simply by larger models giving Ridge a higher-dimensional map. **(3)** The fitted linear components remain broadly anisotropic, yet most of their mKNN lift is carried by a comparatively small singular subspace. Secondary controls—PCA bottlenecks, sparse SAE [Gao et al., 2025]/BSF [Fel et al., 2026] representations, a same-dataset cross-model symmetry check, and an unpaired nonlinear bottleneck—are reported in the appendix.

2 Setup

We use the official UniverseTBD Legacy $\leftrightarrow$ HSC matched embeddings [UniverseTBD et al., 2025] ($n=16,384$ objects) across five families—AstroPTv2 [Smith et al., 2024], ConvNeXtv2 [Woo et al., 2023], DINOv2 [Oquab et al., 2024], ViT [Dosovitskiy et al., 2021], and I-JEPA [Assran et al., 2023]—sixteen size rungs in total (Appendix A). I-JEPA has only two rungs.

For representations A and B ,

$$M_k = \frac{1}{n} \sum_i \frac{|\mathcal{N}_k^A(i) \cap \mathcal{N}_k^B(i)|}{k},$$

with cosine neighbours and self-exclusion [Huh et al., 2024, UniverseTBD et al., 2025]. The primary scale is $k=10$. All primary results use a frozen train/test split ($n_{\text{train}}=13,107$, $n_{\text{test}}=3277$, seed 0) and a **test-only gallery**: neighbours are sought among held-out objects only (chance ≈ 0.0031). Direction is Legacy $\rightarrow$ HSC. We fit an affine Ridge predictor after train-only standardization (StandardScaler on X, Y , then Ridge with $\alpha=1$ and an intercept). Because our geometric diagnostics characterize its linear component A_{eff} in original coordinates, we use “linear recoverability” as shorthand for correspondence exposed by this affine-linear alignment (Appendix E).

3 Linear recoverability scales more strongly than native correspondence

Supervised linear alignment raises both the level and the model-size slope of cross-survey correspondence. Mean held-out mKNN@10 is 0.022 for native correspondence and 0.042 for linear recoverability (alignment gain $G \approx 0.020$; Figure 1a)—roughly a doubling of native overlap and $\sim 13\times$ chance (≈ 0.0031), without claiming downstream astrophysical utility. Family OLS slopes of mKNN versus $\log_{10} P$ steepen under Ridge:

$$\Delta\beta_F = \beta_{\text{Dense+Ridge},F} - \beta_{\text{Dense},F}$$

is positive in all five families, giving

$$T = \frac{1}{5} \sum_F \Delta\beta_F = 0.00721.$$

We average family slope differences equally so that the architecture family—not the number of published checkpoints within a family—is the unit of replication (Appendix A). The amplification is positive in all five architecture families ($p_{\text{sign}}=0.031$), remains positive under held-out-object bootstrap (95% CI [0.0034, 0.0111]), and exceeds a synchronized correspondence-shuffle refit null ($p_{\text{perm}} \approx 0.005$; Appendix A). It remains positive across neighbourhood sizes $k \in \{5, 10, 25, 50\}$; the reverse HSC $\rightarrow$ Legacy map also yields $T_{H\rightarrow L} > 0$, though with smaller magnitude and one negative family (Appendix B).

The same pattern survives at fixed representation dimension. Using *independent* survey PCAs at rank $r=256$ (Appendix C), native $M_{\text{indPCA256}}$ and supervised $M_{\text{indPCA256+Ridge}}$ share the same 256-dimensional features. Defining

$$\Delta\beta_{F,256} = \beta_{\text{indPCA256+Ridge},F} - \beta_{\text{indPCA256},F}, \quad T_{256} = \frac{1}{5} \sum_F \Delta\beta_{F,256} = 0.00864,$$

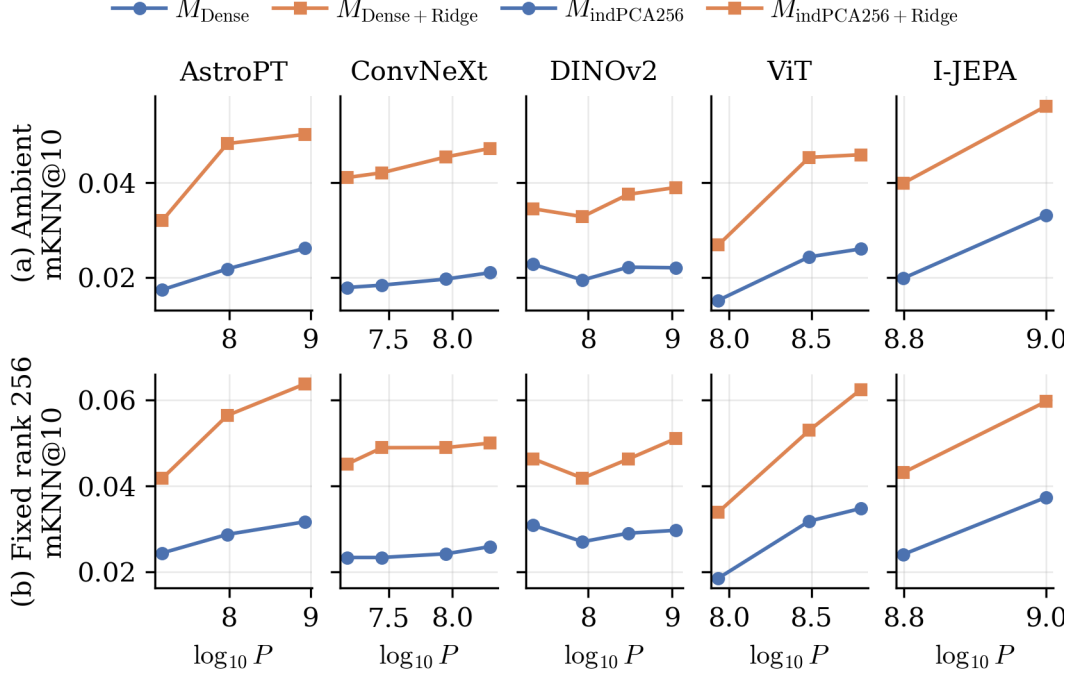


Figure 1: **(a)** Held-out mKNN@10 versus $\log_{10} P$: native M_{native} vs. linear M_{linear} . **(b)** Fixed independent-PCA rank 256. Supervised alignment steepens scaling in both settings ($T > 0$, $T_{256} > 0$).

we find $\Delta\beta_{F,256} > 0$ in all five families ($T_{256}^{(-I)} = 0.00673$ excluding I-JEPA). The effect persists when alignment is restricted to the same 256-dimensional representation space, so it cannot be explained simply by larger models giving Ridge a higher-dimensional map (Figure 1b).

4 Recoverability is concentrated within broadly anisotropic maps

Because scaled-orthogonal maps $x \mapsto cQx$ preserve pairwise cosine similarities, they cannot change cosine mKNN; the observed gain must depend on deformation outside this class (and, for the full affine map, potentially translation). We characterize the learned linear component $A_{\text{eff}} = \text{diag}(\sigma_y)W\text{diag}(1/\sigma_x)$ in original coordinates ($W = \text{Ridge coef}_-$; Appendix E).

Writing $A_{\text{eff}} = U\Sigma V^T$, the log-singular-value dispersion $A_{\log} = \text{std}_i(\log \sigma_i)$ is 1.76–3.15 across rungs, so alignment requires substantial direction-dependent stretch rather than rotation/reflection and global rescaling alone. Defining centered log-stretches $a_i = \log \sigma_i - \frac{1}{d} \sum_j \log \sigma_j$ and anisotropy masses a_i^2 , let f_{90}^{anis} be the fraction of singular directions required for 90% of $\sum a_i^2$. Across the 16 maps, f_{90}^{anis} is 0.411–0.490 (median 0.427): departure from uniform scaling is spread across a substantial fraction of the representation (Figure 2a).

Yet the deformation relevant to neighbourhood recovery is much more concentrated. Truncating A_{eff} along singular directions without refitting recovers 90% of the Dense+Ridge mKNN lift $G = M_{\text{Dense+Ridge}} - M_{\text{native}}$ using only 2.1–17% of directions (median 4.8%; Figure 2b). Spectral anisotropy and functional recoverability therefore have sharply different effective dimensionalities.

This contrast does not itself show a compelling size trend: f_{90}^{anis} decreases slightly in four of five families, but changes are only a few percentage points and the pooled association with $\log_{10} P$ is weak (descriptive Spearman $\rho=0.13$, $p=0.64$, $n=16$); $A_{\log}$ and f_{90}^{anis} are likewise weakly associated ($\rho=0.39$, $p=0.13$; Appendix E). Scaling therefore strengthens linear recoverability without corresponding simplification of the native metric geometry, while the part of A_{eff} that matters for cosine-neighbour recovery remains functionally low-dimensional.

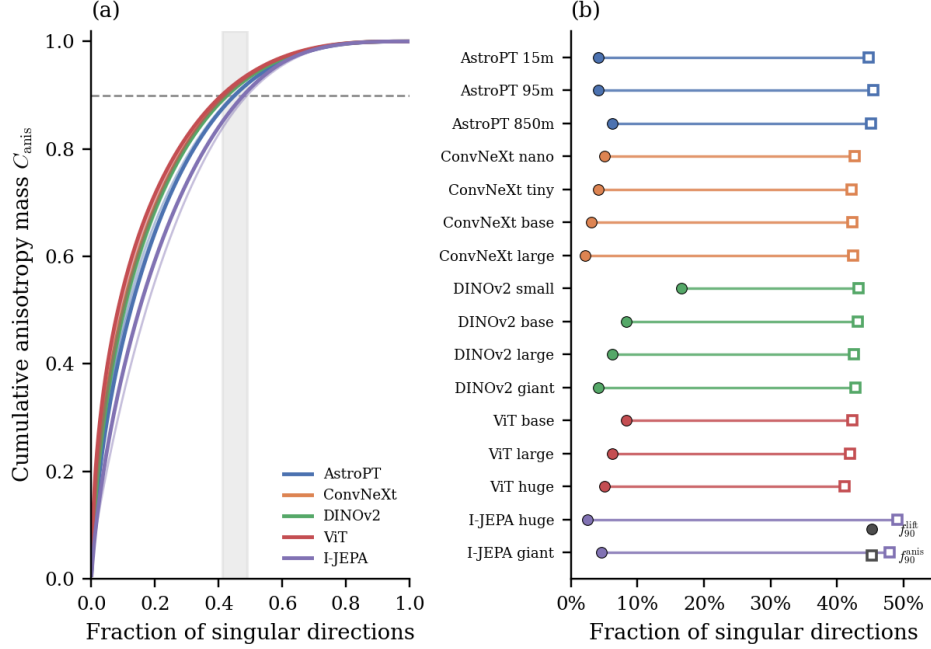


Figure 2: **Broad anisotropy but concentrated functional transfer.** (a) Cumulative anisotropy mass from centered log-singular-value deviations; 90% of this mass requires 41–49% of singular directions (shaded). (b) Per-map fraction of singular directions recovering 90% of Dense+Ridge mKNN lift (f_{90}^{lift} ; filled) versus 90% of anisotropy mass (f_{90}^{anis} ; open). Functional recoverability is substantially more concentrated (median 4.8%) than the anisotropic deformation (median 42.7%). These are properties of the fitted alignment map, not intrinsic representation rank.

5 Discussion and conclusion

Apparent representational convergence depends on the permitted transformation class. Native cosine correspondence grows only weakly with model size, and scaled-orthogonal maps leave cosine neighbourhoods unchanged. At $\mathcal{T}_{\text{affine}}$, supervised recoverability is clearly amplified ($T=0.00721$; 5/5) and survives fixed independent-PCA dimension ($T_{256}=0.00864$; 5/5). Translation ablation leaves most absolute gain and slope amplification intact, so the effect is primarily carried by anisotropic A_{eff} (Appendix E). The geometry of the fitted maps further separates native similarity from recoverability: although the effective linear transformations are broadly anisotropic—a median 43% of singular directions account for 90% of their departure from uniform scaling—their contribution to neighbourhood recovery is much more concentrated (median 4.8% of directions recovers 90% of the Dense+Ridge mKNN lift). Thus linear recoverability can be functionally low-dimensional even when the alignment map as a whole is not a low-dimensional deformation. Reverse amplification is weaker ($T_{H \rightarrow L}=0.00447$); a same-observation control retains $R_{A \rightarrow B} \neq R_{B \rightarrow A}$ (Appendix B,F). Limitations: paired Ridge supervision, five families, one astronomy setting, frozen split; T is stable across Ridge α and k (Appendix A,B). Scaling strengthens supervised linear recoverability more clearly than native cosine similarity.

References

- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499.
- Arjun Dey, David J. Schlegel, Dustin Lang, Robert Blum, Kaylan Burleigh, Xiaohui Fan, Joseph R. Findlay, Doug Finkbeiner, David Herrera, Stéphanie Juneau, et al. Overview of the DESI legacy imaging surveys. *The Astronomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.

126 Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner,
127 Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An
128 image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on*
129 *Learning Representations*, 2021.

130 Thomas Fel, Matthew Kowal, Mozes Jacobs, Dron Hazra, Usha Bhalla, Lee Sharkey, Lucius Bushnaq, Satchel
131 Grant, Tal Haklay, Thomas Icard, Can Rager, Michael Pearce, Daniel Wurgaft, Aiden Swann, Fenil Doshi,
132 Siddharth Boppana, Curt Tigges, Nick Cammarata, Thomas Serre, Vasudev Shyam, Owen Lewis, Thomas
133 McGrath, Jack Merullo, Ekdeep Singh Lubana, and Atticus Geiger. Structuring sparsity: Block-sparse
134 featurizers capture visual concept manifolds. *arXiv preprint arXiv:2606.25234*, 2026.

135 Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan
136 Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning*
137 *Representations*, 2025. URL <https://arxiv.org/abs/2406.04093>.

138 Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The Platonic representation
139 hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan
140 Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine*
141 *Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 20617–20642. PMLR, 2024.
142 URL <https://proceedings.mlr.press/v235/huh24a.html>.

143 Rishi Jha, Collin Zhang, Vitaly Shmatikov, and John X. Morris. Harnessing the universal geometry of embeddings.
144 In *Advances in Neural Information Processing Systems*, 2025. URL [https://proceedings.neurips.cc/
145 paper_files/paper/2025/hash/4175dee33d6145cb8f0323703d138a53-Abstract-Conference.
146 html](https://proceedings.neurips.cc/paper_files/paper/2025/hash/4175dee33d6145cb8f0323703d138a53-Abstract-Conference.html).

147 Guillaume Lample, Alexis Conneau, Marc’Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. Word
148 translation without parallel data. In *International Conference on Learning Representations*, 2018. URL
149 <https://openreview.net/forum?id=H196sainb>.

150 Tomas Mikolov, Quoc V. Le, and Ilya Sutskever. Exploiting similarities among languages for machine translation.
151 *arXiv preprint arXiv:1309.4168*, 2013.

152 Satoshi Miyazaki, Yutaka Komiyama, Satoshi Kawanomoto, Yukiko Fujimori, Takashi Hamana, Hisanori
153 Hayashi, Hiroki Heiguchi, Takashi Hattori, Shiang-Yu Huang, Masatoshi Imanishi, et al. Hyper Suprime-
154 Cam: System design and verification of image quality. *Publications of the Astronomical Society of Japan*, 70
155 (SP1):S1, 2018. doi: 10.1093/pasj/psx063.

156 Luca Moschella, Valentino Maiorca, Marco Fumero, Antonio Norelli, Francesco Locatello, and Emanuele
157 Rodolà. Relative representations enable zero-shot latent space communication. In *International Conference*
158 *on Learning Representations*, 2023. URL <https://openreview.net/forum?id=Src-nwieGJ>.

159 Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre
160 Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech
161 Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel
162 Synnaeve, Hu Xu, Hervé Jégou, Julien Labatut, Armand Joulin, and Piotr Bojanowski.
163 DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*,
164 2024.

165 Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large
166 observation models for astronomy. In *ICML 2024 AI for Science Workshop*, 2024.

167 UniverseTBD, Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The Platonic universe: Do
168 foundation models see the same sky? *arXiv preprint arXiv:2509.19453*, 2025. doi: 10.48550/arXiv.2509.
169 19453. Accepted to the NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences.

170 Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie.
171 ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. In *Proceedings of the*
172 *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16133–16142, 2023.

173 A Protocol and statistical robustness

174 Approximate parameter counts (10^6): AstroPTv2 15/95/850; ConvNeXtv2 nano/tiny/base/large
175 15/28/89/198; DINOv2 small/base/large/giant 22/86/300/1100; ViT base/large/huge 86/307/632;
176 I-JEPA huge/giant 630/1000.

The primary statistic $T = \frac{1}{5} \sum_F \Delta\beta_F$ averages family slope differences equally so that architecture family—not checkpoint count—is the replication unit. The forward effect is positive in all five families ($p_{\text{sign}}=0.03125$), exceeds a synchronized correspondence-shuffle refit null ($B=200$, $p=1/201 \approx 0.005$), and remains positive under held-out-object bootstrap (95% CI $[0.0034, 0.0111]$); excluding I-JEPA gives $T_{-I}=0.00537$ with four/four families positive. The result is also stable across $\alpha \in \{0.01, 0.1, 1, 10, 100\}$, with $T(\alpha) \in [0.00721, 0.00848]$ and 5/5 positive families throughout.

B Direction and neighbourhood-size robustness

All checks reuse the frozen train/test split, Ridge recipe, and test-only gallery.

Reverse direction. HSC→Legacy yields $T_{H \rightarrow L}=0.00447$ with 4/5 families positive (AstroPT negative). Excluding I-JEPA, $T_{H \rightarrow L}^{(-I)}=0.00103$. Reverse amplification is present but weaker and family-heterogeneous; it is not a second primary claim.

Neighbourhood size. Aggregate forward amplification is stable across $k \in \{5, 10, 25, 50\}$: $T(k) \in \{0.00782, 0.00721, 0.00810, 0.00869\}$, with 5/5 positive families at $k=5, 10$ and 4/5 at $k=25, 50$.

Pooled interaction. On a $\text{rung} \times \text{method}$ table, a family-fixed-effects OLS specification $M = \alpha_F + \gamma_F \log_{10} P + \delta_F R + \theta(R \log_{10} P)$ gives the same positive interaction direction ($\theta=0.00451$), providing a descriptive robustness check under a different weighting. We do not treat ordinary IID OLS uncertainties for this pooled model as primary inference, because Dense/Ridge observations are paired by rung. Equal-family T , the family sign test, the synchronized shuffle-refit null, and the object bootstrap remain the primary evidence.

C Secondary representation controls

Independent PCA+Ridge raises absolute recoverability at mid/high ranks (means $\approx 0.044/0.050/0.055$ at 128/256/512); the fixed-rank 256 result is reported in the main text. A pooled train-only PCA basis without pairing leaves native mKNN near Dense (≈ 0.024 at rank 256), showing that dimensional reduction alone does not substantially increase native correspondence. PCA does not yield a consistent additional scale interaction beyond Dense+Ridge.

Under matched Ridge supervision, SAE and BSF underperform Dense+Ridge by 0.0085 and 0.0062 mKNN on average, respectively; family-bootstrap 95% CIs are $[-0.0120, -0.0043]$ and $[-0.0104, -0.0007]$. Sparse coding therefore provides no additional cross-survey correspondence once alignment supervision is controlled. SAE protocol: width $F=2048$, seed 0, TopK from available checkpoints, train-only Ridge/IDF, fixed Legacy→HSC [Gao et al., 2025]; BSF uses Vanilla signed codes with the same Ridge recipe [Fel et al., 2026]. Inference-time TopK sensitivity is reported in Appendix D.

Global Ridge residuals remain weakly locally structured in source space ($k \in \{10, 25, 50\}$; all 16×3 residual-permutation tests give $p=1/201$), but absolute residual cosine similarity is ≈ 0.01 and mean relative held-out MSE reduction from a patch intercept is $\ll 0.1\%$; we do not fit local maps.

D SAE TopK sensitivity

Trained checkpoints use heterogeneous TopK in 18–23 with no common training-time K across all 16 rungs. To test whether the sparse-control result depends on sparsity level, we therefore repeat the frozen Legacy→HSC protocol under *inference-time* TopK overrides $K \in \{10, 20, 40\}$ on the preferred $F=2048$ seed-0 checkpoints, holding dictionary width, split, Ridge fitting, mapping direction, IDF, and the test-only gallery fixed. Across this range the mean residual $M_{\text{SAE+Ridge}} - M_{\text{Dense+Ridge}}$ stays negative ($-0.0159, -0.0085, -0.0077$), family-bootstrap 95% CIs exclude zero, and the equal-family scaling interaction $T_{\text{SAE}, K}$ remains non-positive ($-0.0133, -0.0099, -0.0079$; at most one family positive). Thus the sparse result is not specific to the original TopK choice: across the tested inference sparsity range, SAE does not improve correspondence over the matched Dense+Ridge baseline and does not yield a consistently stronger size trend (Figure 3).

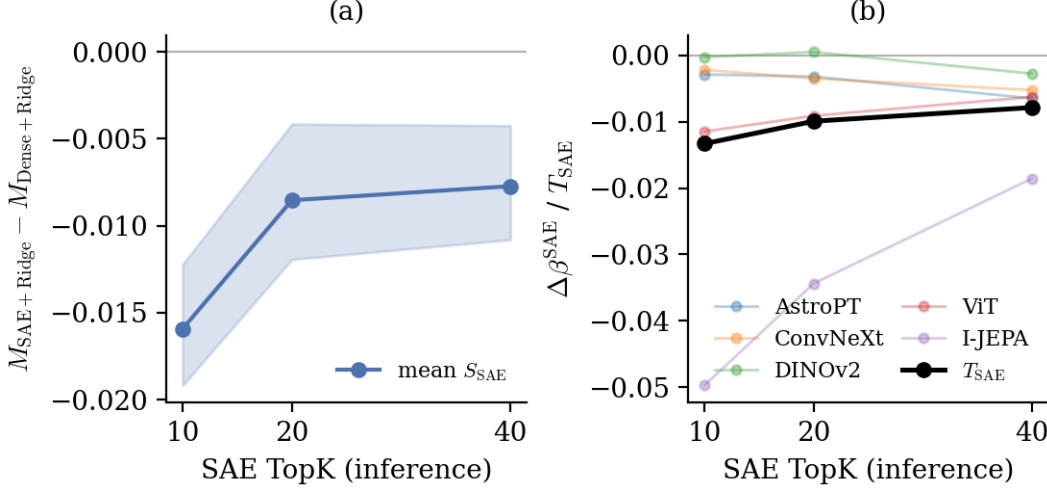


Figure 3: **(a)** Mean SAE residual versus Dense+Ridge under inference-time TopK $K \in \{10, 20, 40\}$ (family-bootstrap 95% CI). **(b)** Equal-family scaling interaction $T_{\text{SAE},K}$ with per-family $\Delta\beta_{F,K}^{\text{SAE}}$ (faint). Both absolute level and size interaction remain non-positive across K .

223 E Geometry details

224 After train-only StandardScaler, Ridge yields W and an intercept in standardized coordinates. The
 225 full affine map in original coordinates is $\hat{y} = A_{\text{eff}}x + b_{\text{eff}}$ with $A_{\text{eff}} = \text{diag}(\sigma_y)W\text{diag}(1/\sigma_x)$ and
 226 $b_{\text{eff}} = \mu_y - A_{\text{eff}}\mu_x + D_y b_{\text{std}}$. All spectral diagnostics below characterize the *linear component* A_{eff} ,
 227 not b_{eff} .

228 Writing $A_{\text{eff}} = U\Sigma V^\top$, we distinguish several quantities. *Magnitude of anisotropy.* $A_{\log} =$
 229 $\text{std}_i(\log \sigma_i)$ measures overall departure from uniform scaling (1.76–3.15 across rungs). *Concen-*
 230 *tration of anisotropy.* Centered log-stretches $a_i = \log \sigma_i - \text{mean}_j \log \sigma_j$ yield anisotropy masses
 231 $m_i = a_i^2$; f_{90}^{anis} is the fraction of singular directions required for 90% of $\sum m_i$ (0.411–0.490; median
 232 0.427). *Functional concentration of lift.* Truncating A_{eff} along singular directions *without*
 233 *refitting*, on the same held-out objects and test-only gallery, and defining $G_r = M_r - M_{\text{native}}$
 234 relative to the full-map lift G_{full} , the smallest evaluated rank with $G_r/G_{\text{full}} \geq 0.9$ uses 2.1–
 235 17% of directions (median 4.8%). Anisotropy concentration and functional lift concentration
 236 are therefore distinct measurements of the fitted map. *Distance from the cosine-preserving class.*
 237 $D_{\text{sim}} = \min_{c,Q^\top Q=I} \|A_{\text{eff}} - cQ\|_F / \|A_{\text{eff}}\|_F$ is 0.891–1.000; because scaled-orthogonal maps
 238 cannot change cosine mKNN, a large D_{sim} characterizes the type of deformation required rather
 239 than independently explaining the gain. *Local stretch.* On Legacy test k_{edge} -NN edges δ_{ij} ,
 240 $s_{ij} = \log(\|A_{\text{eff}}\delta_{ij}\|_2 / \|\delta_{ij}\|_2)$ with $\sigma_{\text{local}} = \text{std}(s_{ij})$ is 0.157–0.310 ($k_{\text{edge}}=10$), so anisotropy
 241 also affects locally occupied held-out directions; σ_{local} decreases with size in four of five families
 242 (suggestive, not conclusive). Secondary spectral-energy diagnostics (H_{norm}) likewise show non-flat
 243 maps. Scale trends of $A_{\log}$, f_{90}^{anis} , and σ_{local} are shown in Figure 4; pooled Spearman associations
 244 with $\log_{10} P$ are weak for f_{90}^{anis} ($\rho=0.13$, $p=0.64$).

245 Because cosine mKNN is not translation-invariant, we ablate b_{eff} without refitting: for each rung we
 246 compare native $M(X, Y)$, full affine $M(A_{\text{eff}}X + b_{\text{eff}}, Y)$, and linear-only $M(A_{\text{eff}}X, Y)$ on the same
 247 test queries/gallery. Affine reconstruction matches the sklearn pipeline to $\max |\text{err}| \leq 3.5 \times 10^{-5}$.
 248 Across the 16 rungs the fraction of affine gain retained by the linear component is $\eta_A = \text{median } 0.990$
 249 (mean 0.985; range [0.920, 1.048]). Family slope amplification without translation is $T_A=0.00854$
 250 (5/5 positive) versus $T_{\text{affine}}=0.00721$, so $T_A/T_{\text{affine}}=1.184$ (excluding I-JEPA, $T_A^{(-I)}=0.00570$).
 251 A train-mean-centered diagnostic likewise preserves amplification ($T_{\text{centered}}=0.01360$; Figure 5).
 252 Thus the headline scaling effect is primarily attributable to anisotropic linear deformation rather than
 253 affine translation.

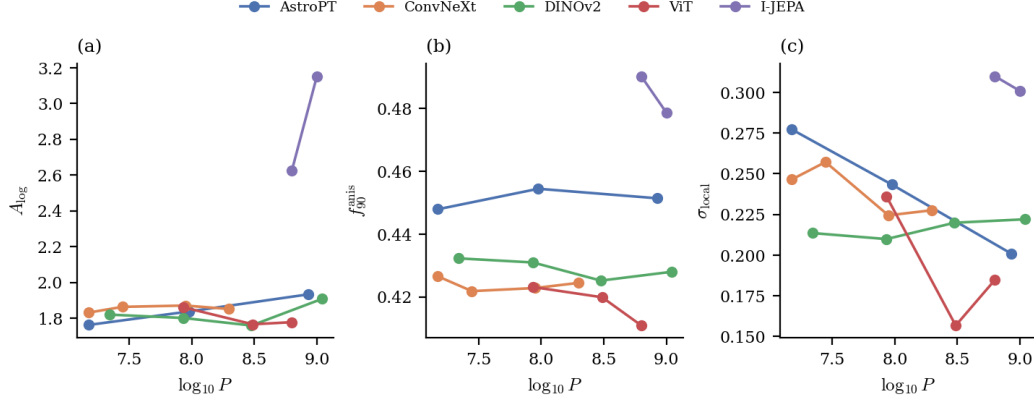


Figure 4: Within-family trends versus $\log_{10} P$: **(a)** anisotropy magnitude $A_{\log}$; **(b)** anisotropy concentration f_{90}^{anis} ; **(c)** local stretch dispersion σ_{local} on held-out Legacy edges. Points are connected only within architecture families.

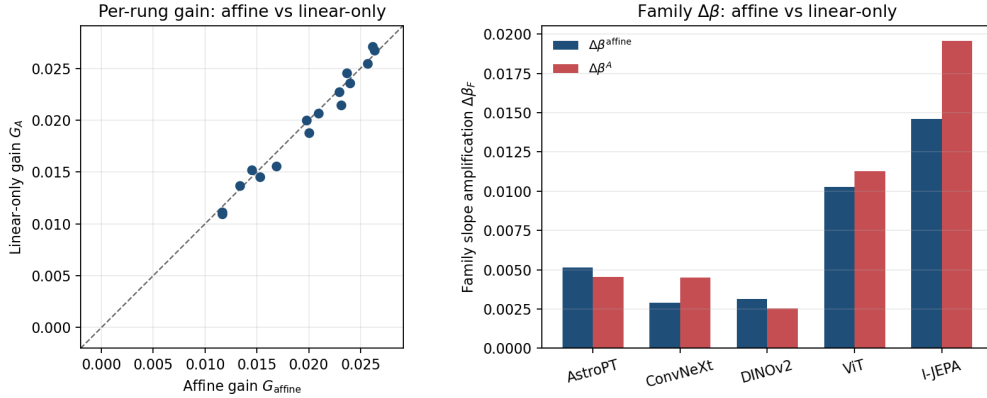


Figure 5: **(a)** Per-rung Dense+Ridge mKNN gain with the full affine map versus the linear component alone ($y=x$ dashed). **(b)** Family slope amplifications $\Delta\beta_F$ for affine and linear-only conditions. Removing b_{eff} retains essentially all absolute gain and the model-size amplification.

254 F Same-dataset cross-model control

255 Cross-survey directionality may reflect survey-information differences, so same-Legacy cross-model
 256 alignment tests whether asymmetry persists when the observation channel is fixed. We evaluate all
 257 101 valid unordered cross-family pairs among 16 Legacy models (train-only independent PCA256
 258 per model; Ridge $\alpha=1$; mKNN@10; test-only gallery; frozen 80/20 split; both directions fit indepen-
 259 dently). For unordered pair $\{A, B\}$,

$$G_{\text{sym}} = \frac{1}{2}(G_{A \rightarrow B} + G_{B \rightarrow A}), \quad A_{\text{dir}} = |G_{A \rightarrow B} - G_{B \rightarrow A}|,$$

260 where $G_{A \rightarrow B} = M_{A \rightarrow B}^{\text{Ridge}} - M_{\text{native}}(A, B)$ and likewise for $G_{B \rightarrow A}$.

261 Mean native mKNN is 0.050 (chance ≈ 0.00305). Ridge increases correspondence in both directions
 262 for 101/101 pairs (mean $G_{\text{sym}}=0.060$; mean $A_{\text{dir}}=0.036$; mean relative asymmetry $A_{\text{rel}} \approx 0.33$).
 263 Mean directional gain is similar for larger $\rightarrow$ smaller (≈ 0.059) and smaller $\rightarrow$ larger (≈ 0.062) maps,
 264 so there is no systematic larger-to-smaller advantage. Descriptively, both mutual recoverability
 265 and directional asymmetry increase with the scale of the smaller model ($\beta_{\min}(G_{\text{sym}})=+0.026$,
 266 $\beta_{\min}(A_{\text{dir}})=+0.017$); pair-level trends are not treated inferentially because model endpoints recur
 267 across pairs.

268 **G Unpaired nonlinear control**

269 An unpaired DualEncoder shared bottleneck (model-specific MLPs into $Z=256$; reconstruction,
270 RBF-MMD, cycle, and Gram terms; no paired cross-survey training objects) yields held-out relational
271 mKNN@10 of 0.012–0.023 versus chance ≈ 0.0035 , but size dependence is non-monotonic on the
272 available ConvNeXt and DINOv2 ladders.